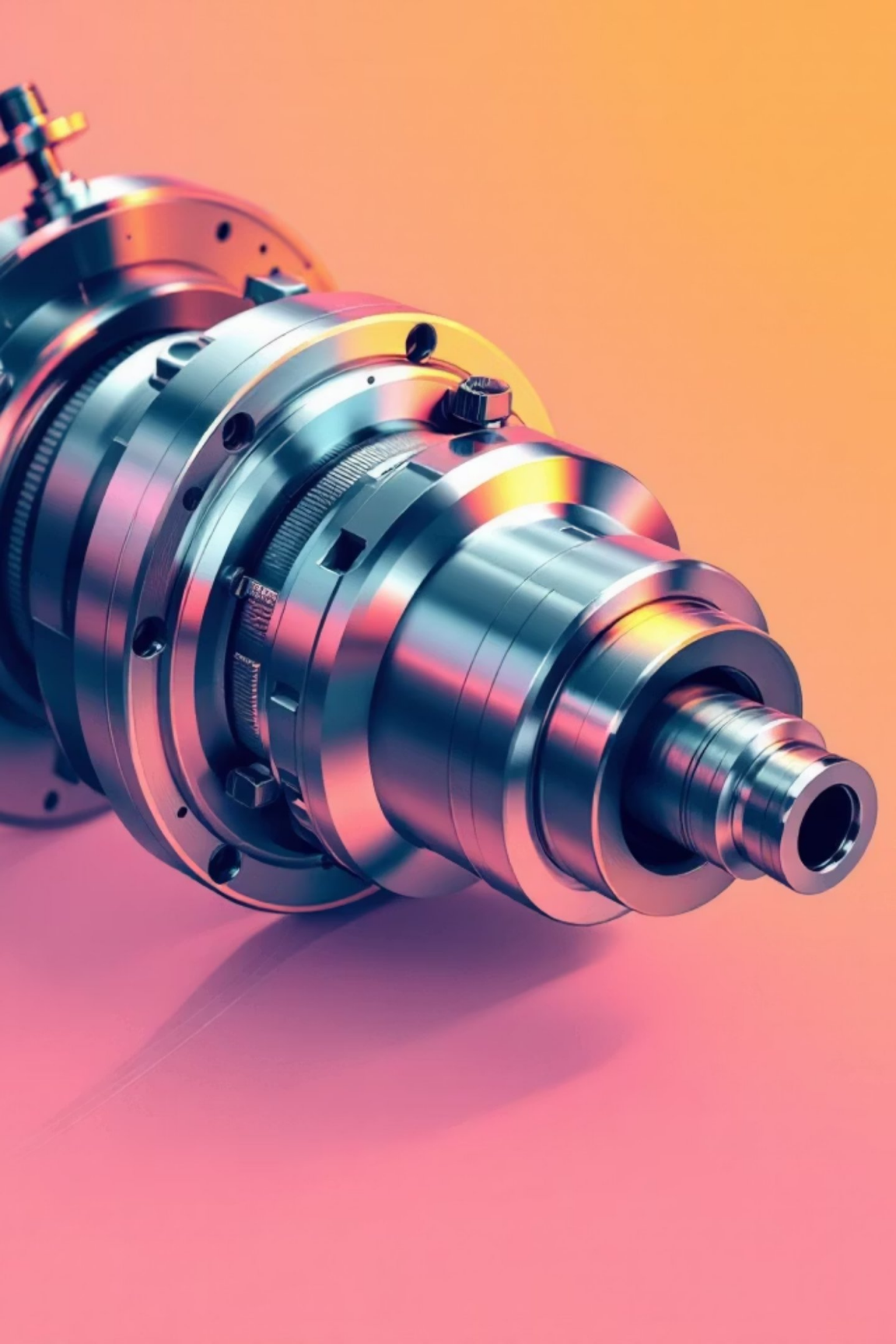




Schmidt Coupling Mechanism

Design and Analysis using **Autodesk Fusion** — An engineering project showcase exploring the 3D modeling, working principle, and industrial applications of a high-performance parallel shaft coupling

MECHANICAL DESIGN FUSION 360 KINEMATICS



What is a Schmidt Coupling?

Invented by **Walter Schmidt** in the 1960s, the Schmidt Coupling is a zero-backlash flexible shaft coupling designed to transmit torque between two parallel shafts with variable offset — all while maintaining **constant angular velocity**.

Origin

Developed in Germany for precision industrial drive systems requiring smooth torque transmission.

Purpose

Connects parallel shafts with lateral offset without introducing speed variation or backlash.

Classification

A type of **Oldham coupling variant** using a three-disk linkage system for superior rigidity.

Design Methodology in Autodesk Fusion

The Schmidt Coupling was fully modeled in **Autodesk Fusion 360** using a structured parametric workflow — from 2D sketches to a fully constrained assembly ready for simulation.

Extrude & Feature

Create 3D solids with tolerances.

Assembly & Joints

Apply rigid and revolute constraints.

Sketch Profiles

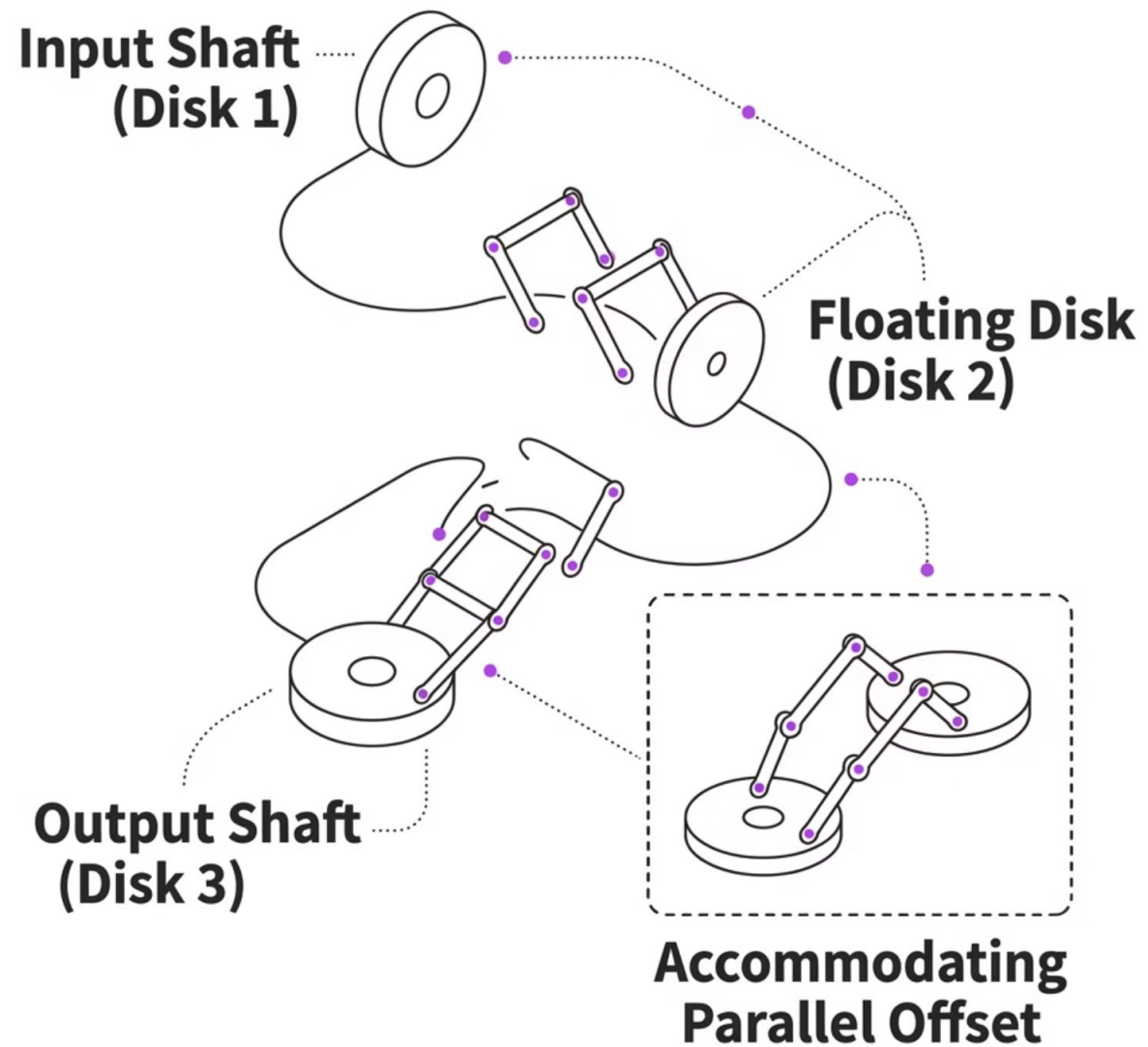
Define 2D disks and linkage geometry.

Motion Study & FEA

Run kinematic and stress validation.

Parametric constraints and Fusion's **As-Built Joints** enabled rapid design iteration and accurate motion simulation without manual rework.

Working Principle



Constant Velocity Through Linkage Geometry

The coupling uses a four-bar parallelogram linkage formed by pins and linkages connecting the three disks. As the input shaft rotates, the intermediate disk translates in a circular path while remaining parallel to both shafts.

- Input and output shafts rotate at **identical RPM**
- Lateral offset is absorbed by the floating disk
- No speed fluctuation — ideal for precision machinery

Key Components



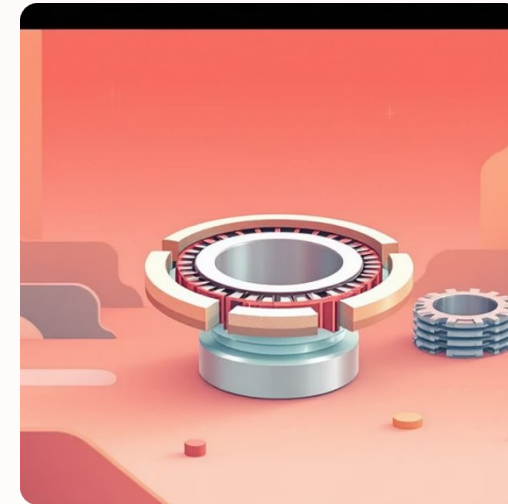
Three-Disk Arrangement

Input disk, floating intermediate disk, and output disk form the core kinematic chain. All disks remain parallel during operation.



Linkages & Pins

Four hardened steel pins connect the disks via rigid linkages, forming the parallelogram mechanism that governs motion transfer.



Needle Bearings

Needle roller bearings at each pin joint minimize friction and wear, enabling high-speed operation with long service life.

Technical Advantages



Zero Backlash

Preloaded pin joints eliminate clearance, delivering precise positional accuracy under reversing torque loads.



High Torque Capacity

Multi-pin load distribution allows the coupling to handle significant torque without deformation or failure.



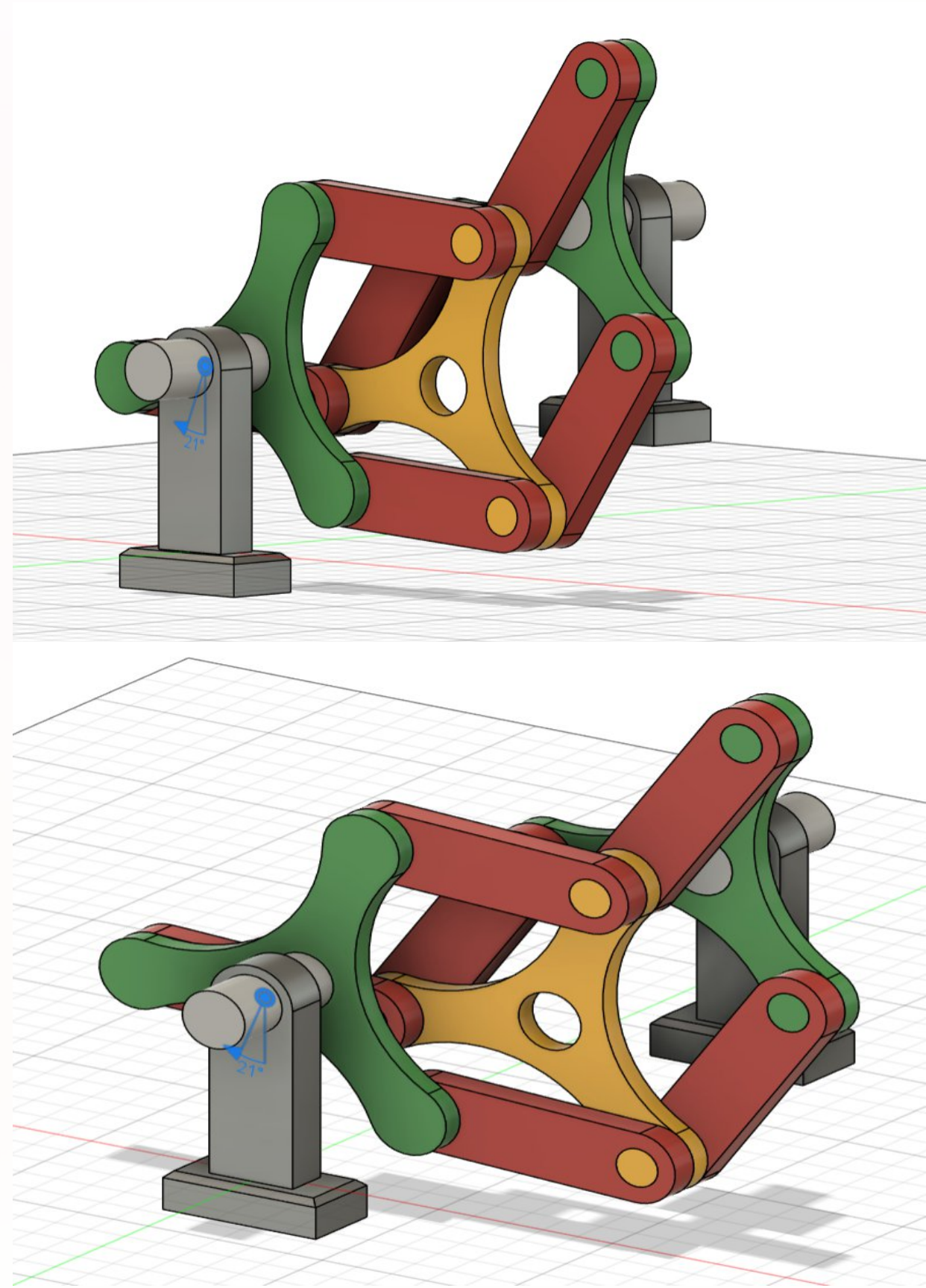
Compact Footprint

Axially short design fits into tight spaces where universal joints or gear couplings cannot be accommodated.



Constant Velocity

Unlike flexible couplings, the Schmidt design guarantees a **1:1 speed ratio** regardless of shaft offset angle.



Industrial Applications

The Schmidt Coupling's ability to handle parallel offset with zero backlash makes it the preferred choice across several heavy-industry sectors.



Printing Presses

Drives synchronized roller trains where precise registration and constant speed are critical to print quality.



Roll Forming Machines

Transmits power to forming rolls that must be repositioned frequently while maintaining synchronized rotation.



Paper Processing

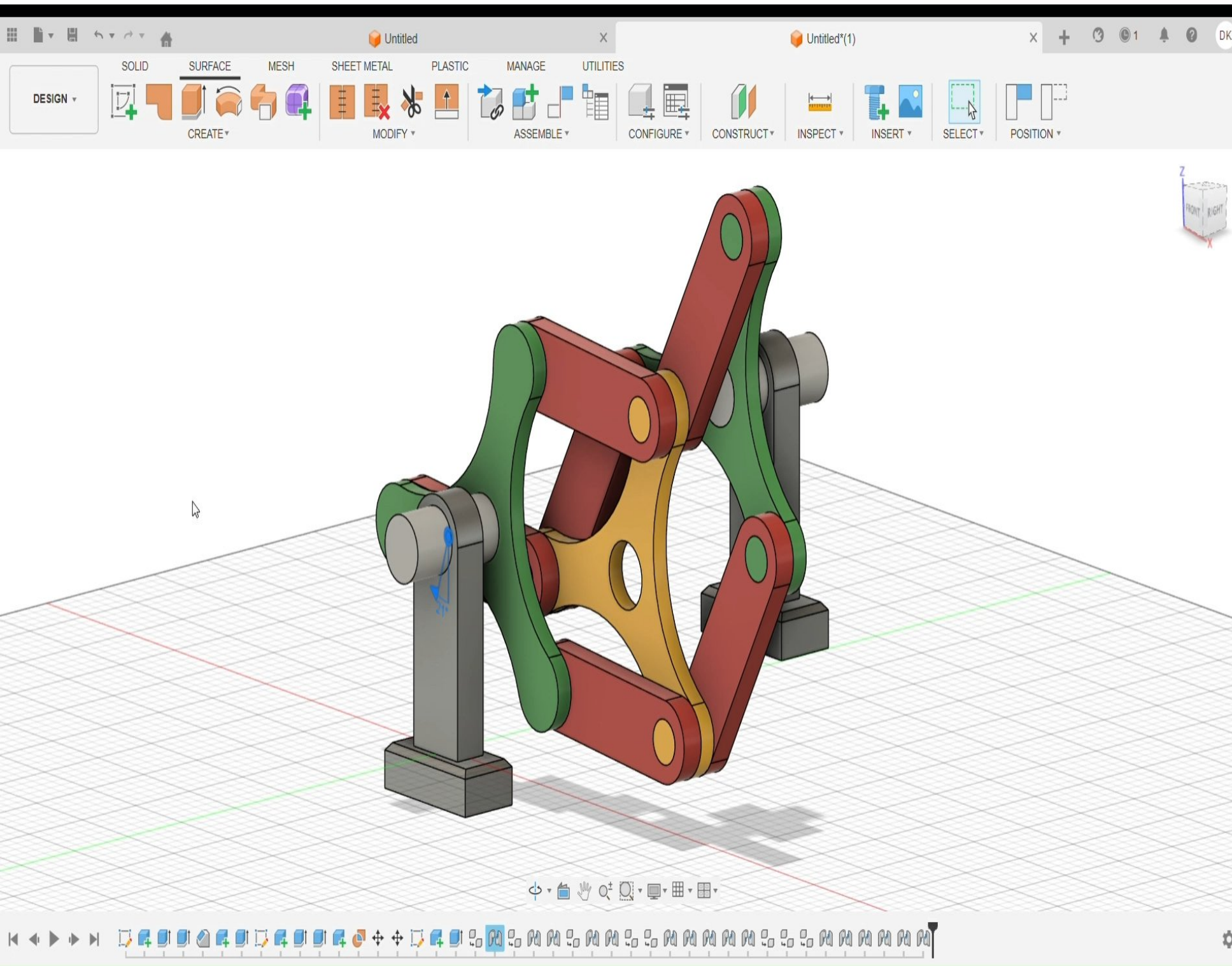
Connects drive shafts in calender and coating machines with adjustable lateral spacing for different web widths.



Textile Industry

Used in looms and winding machines requiring smooth, vibration-free torque transmission at high RPM.

Simulation & Analysis Results



Motion Study

A **360° rotational drive joint** was applied to the input shaft. The motion study confirmed constant output velocity and validated the parallelogram linkage geometry with zero positional error.

Static Stress Analysis (FEA)

Structural simulation under rated torque revealed maximum von Mises stress well below the yield strength of steel (AISI 1045), confirming a safe factor of safety > 2.0.

- ❏ No interference or binding detected across the full range of shaft offset positions.



Conclusion & Future Scope

Project Summary

This project demonstrated end-to-end design of a Schmidt Coupling in Fusion 360 — from parametric sketching and assembly to kinematic validation and FEA. The coupling meets all functional requirements for zero-backlash parallel shaft drive applications.

Future Scope

- Optimize disk thickness and pin diameter for weight reduction
- Explore additive manufacturing for lightweight prototype builds
- Perform dynamic fatigue analysis for lifecycle prediction
- Design a variable-offset adjustment mechanism for field use